Worksheet 3-1 A The Chain Rule #1
AP Calculus Name_____

Directions: For 1 – 16, find the derivative of the following functions.

1. $f(x) = \sqrt[3]{x^3 - 4}$

2. $g(\theta) = \sin(\theta^2)$

3. $h(x) = \cos^2(x)$

4. $f(x) = \ln|x^2 + 5|$

5. $y = \dfrac{1}{x^2 - 3x + 1}$

6. $k(t) = e^{4t}$

7. $f(x) = (x^3 - 2x + 5)^4$

8. $f(x) = \sqrt{\tan(x)}$

9. $k(x) = \dfrac{1}{e^{3x}}$

10. $f(x) = e^{\sin(x)}$

11. $f(x) = \sin^2(x)$

12. $f(x) = \ln|2x^3 - 4x + 7|$

13. $y = \cos(\ln|x|)$

14. $y = \ln|\tan(3x)|$

15. $f(x) = \dfrac{-3}{x^2 - 3x + 6}$

16. $g(x) = f(2x)e^{-4x}$

Directions: For 17 – 19, find the equation of the tangent line at the given *x*-value.

17. $f(x) = (1 + \tan x)^{3/2}$ at $x = 0$

18. $g(x) = \dfrac{1}{(3 - 2x)^2}$ at $x = 1$

19. $f(x) = e^{3x^4}$ at $x = 0$